Lecture 2: Cardinality and Countability

Mathematical Analysis

1 Cardinality

Definition 1.1 (Cardinality). [def:cardinality] The *cardinality* of a set A, denoted by $|A|$, is the number of elements in A.

Definition 1.2 (Cardinal Inequality). [def:cardinal-inequality] For sets A and B,

$$|A| \leq |B| \iff \exists f : A \to B \text{ such that } f \text{ is injective.}$$

We write $|A| \geq |B|$ if $|B| \leq |A|$. Furthermore,

$$|A| < |B| \iff |A| \leq |B| \text{ and } |A| \neq |B|,$$

and $|A| > |B|$ if $|B| < |A|$.

Definition 1.3 (Equinumerous Sets). [def:same-cardinality] Sets A and B are *equinumerous*, written $A \sim B$, if and only if their cardinalities are equal:

$$A \sim B \iff |A| = |B|.$$

Theorem 1.1 (Bijection Characterization). [thm:bijection-cardinality]

$$|A| = |B| \iff \exists f : A \to B \textit{ such that } f \textit{ is bijective.}$$

Lemma 1.1 (Finite Subsets of Equal Cardinality). [lem:finite-subset-equal-cardinality] *Let $A \subseteq B$, where B is finite. If $|A| = |B|$, then $A = B$.*

Proof. If $A \neq B$, choose $b \in B \setminus A$. Then $A \subseteq B \setminus \{b\}$, so $|A| \leq |B| - 1 < |B|$, a contradiction. □

Lemma 1.2 (Mapping Characterization). [lem:cardinal-order-mappings] *Assume $B \neq \emptyset$ and the Axiom of Choice. Then*

$$\begin{aligned}
|A| \leq |B| &\iff \exists f : A \to B \textit{ injective,} \\
|A| < |B| &\iff \exists f : A \to B \textit{ injective but not bijective,} \\
|A| \geq |B| &\iff \exists f : A \to B \textit{ surjective,} \\
|A| > |B| &\iff \exists f : A \to B \textit{ surjective but not bijective.}
\end{aligned}$$

Definition 1.4 (Equivalence Relation). [def:equivalence-relation] A relation $\sim$ on a set A is called an *equivalence relation* if it satisfies the following properties:

1. Reflexive: $\forall a \in A, a \sim a$
2. Symmetric: $\forall a, b \in A, a \sim b \implies b \sim a$
3. Transitive: $\forall a, b, c \in A, a \sim b, b \sim c \implies a \sim c$

Theorem 1.2 (Cardinality is an Equivalence Relation). [thm:equivalence-relation-cardinality] *The Cardinality relation $\sim$ on the set of all sets is an equivalence relation.*

Example 1.1. $\mathbb{N} \sim \mathbb{N}_1$. Witness: $f : \mathbb{N} \to \mathbb{N}_1, f(n) = n + 1$

Remark 1.1. For infinite sets, the cardinality maybe still the same even if we remove some (or a finite number of) elements from the set.

Theorem 1.3 (Integers and Natural Numbers are Equinumerous). [thm:integers-natural-equinum] $\mathbb{Z} \sim \mathbb{N}$. *Witness:*

$$f = \begin{cases} n/2 & \textit{if } n \textit{ is even} \\ -(n+1)/2 & \textit{if } n \textit{ is odd} \end{cases}$$

2 Power Sets

Definition 2.1 (Power Set)**.** [def:power-set] The *power set* of a set A, denoted by $\mathcal{P}(A)$, is the set of all subsets of A:

$$\mathcal{P}(A) = \{B : B \subseteq A\}.$$

Example 2.1. Let $A = \{1, 2, 3\}$. Then

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, A\}.$$

Theorem 2.1 (Cardinality of Finite Power Set)**.** [thm:cardinality-finite-power-set] *For any finite set A, $|\mathcal{P}(A)| = 2^{|A|}$.*

Lemma 2.1 (Singleton in the Power Set)**.** [lem:singleton-in-power-set] *For every set A,*

$$\forall a \in A, \quad \{a\} \in \mathcal{P}(A).$$

Proof. If $a \in A$, then $\{a\} \subseteq A$. Hence $\{a\} \in \mathcal{P}(A)$. □

Theorem 2.2 (Cantor's Theorem)**.** [thm:cantor-power-set] *There does not exist a surjective function $f : A \to \mathcal{P}(A)$.*

Proof. Assume for contradiction that there exists a surjective function $f : A \to \mathcal{P}(A)$. This means $\forall B \in \mathcal{P}(A), \exists a \in A$ such that $f(a) = B$.
We can seperate all elements in A into two groups:

$$A_1 = \{a \in A : a \notin f(a)\}$$
$$A_2 = \{a \in A : a \in f(a)\}$$

Notice that $A_1 \subseteq A$, so $A_1 \in \mathcal{P}(A)$.
And, since f is surjective, there exists $\alpha \in A$ such that $f(\alpha) = A_1$.
According to PNP Axiom, α must be in either A_1 or A_2.
If $\alpha \in A_1$, then by definition $\alpha \notin f(\alpha) = A_1$, contradicting $\alpha \in A_1$.
If $\alpha \in A_2$, then by definition $\alpha \in f(\alpha) = A_1$. Hence $\alpha \in A_1$, whose definition gives $\alpha \notin f(\alpha)$, also a contradiction.
Both possible cases lead to a contradiction. Therefore, the assumed surjective function $f : A \to \mathcal{P}(A)$ does not exist. □

Corollary 2.1 (Power Set Strictly Larger than Original Set)**.** [cor:power-set-strictly-larger] *For any set A, $|A| < |\mathcal{P}(A)|$.*

Remark 2.1 (Largest Cardinality does not Exist). [rem:largest-cardinality-does-not-exist] There is no largest cardinality for sets since for any set A, $|A| < |\mathcal{P}(A)|$.

3 Images

Definition 3.1 (Image Operator)**.** [def:image] For sets A and B, the *image operator* is the mapping

$$(-)^{\text{img}} : (A \to B) \to \big(\mathcal{P}(A) \to \mathcal{P}(B)\big)$$

defined by

$$f \mapsto (C \mapsto \{f(x) : x \in C\}).$$

Corollary 3.1 (Image of a Subset)**.** [cor:image-of-subset] *For a mapping $f : A \to B$ and a set $C \subseteq A$,*

$$f^{\text{img}}(C) = \{f(x) : x \in C\} \subseteq B.$$

Lemma 3.1 (Monotonicity of Images). [lem:image-monotonicity] *Let $g : X \to Y$ and let $E, F \subseteq X$. Then*

$$E \subseteq F \implies g^{\text{img}}(E) \subseteq g^{\text{img}}(F).$$

Proof. If $y \in g^{\text{img}}(E)$, then $y = g(x)$ for some $x \in E$. Since $E \subseteq F$, we have $x \in F$, and therefore $y \in g^{\text{img}}(F)$. □

Definition 3.2 (Preimage Operator). [def:pre-image] For sets A and B, the *preimage operator* is the mapping

$$(-)^{\text{preimg}} : (A \to B) \to \big(\mathcal{P}(B) \to \mathcal{P}(A)\big)$$

defined by

$$f \mapsto (D \mapsto \{x \in A : f(x) \in D\}).$$

Corollary 3.2 (Preimage of a Subset). [cor:preimage-of-subset] *For a mapping $f : A \to B$ and a set $D \subseteq B$,*

$$f^{\text{preimg}}(D) = \{x \in A : f(x) \in D\} \subseteq A.$$

Lemma 3.2 (Monotonicity of Pre-images). [lem:preimage-monotonicity] *Let $g : X \to Y$ and let $E, F \subseteq Y$. Then*

$$E \subseteq F \implies g^{\text{preimg}}(E) \subseteq g^{\text{preimg}}(F).$$

Remark 3.1 (Image of Pre-image). [rem:image-of-pre-image] In general cases, for a mapping $f : A \to B$,

$$C \subseteq A, \not\Rightarrow f^{\text{preimg}}(f^{\text{img}}(C)) = C$$
$$D \subseteq B, \not\Rightarrow f^{\text{img}}(f^{\text{preimg}}(D)) = D$$

Pf. Counterexamples. Define $f : \mathbb{N} \to \mathbb{B}$ by

$$f(n) = \begin{cases} \text{ff}, & n = 0, \\ \text{tt}, & n \in \mathbb{N}_1. \end{cases}$$

Then

$$f^{\text{preimg}}(f^{\text{img}}(\{1\})) = f^{\text{preimg}}(\{\text{tt}\}) = \mathbb{N}_1 \neq \{1\}.$$

Define $g : \mathbb{B} \to \mathbb{N}$ by

$$g(\text{tt}) = 0, \qquad g(\text{ff}) = 1.$$

Since 2 is not in the range of g,

$$g^{\text{img}}(g^{\text{preimg}}(\{2\})) = g^{\text{img}}(\emptyset) = \emptyset \neq \{2\}.$$

□

Remark 3.2 (What to do when a result is not true?). [rem:how-to-treat-false-results] When a result such as the equalities in remark 3.1 is not true, we may:

- strengthen the assumptions, as in proposition 3.2;
- narrow down the conclusion, as in proposition 3.1.

Proposition 3.1 (Subset Relation for Pre-image of Image). [prop:preimage-image-subset] *In general cases, for a mapping $f : A \to B$,*

$$C \subseteq A, f^{\text{preimg}}(f^{\text{img}}(C)) \supseteq C$$
$$D \subseteq B, f^{\text{img}}(f^{\text{preimg}}(D)) \subseteq D$$

Proposition 3.2 (Pre-image of Image for Bijection Mapping). [prop:bijection-preimage-image] *If $f : A \to B$ is a bijection mapping, then*

$$C \subseteq A, \implies f^{\text{preimg}}(f^{\text{img}}(C)) = C$$
$$D \subseteq B, \implies f^{\text{img}}(f^{\text{preimg}}(D)) = D$$

4 Nested Sequence of Sets

Definition 4.1 (Nested Sequence of Sets). [def:nested-sequence-of-sets] A sequence of sets $(A_n)_{n\in\mathbb{N}}$ is *nested* if

$$\forall n \in \mathbb{N}, \quad A_{n+1} \subseteq A_n.$$

Theorem 4.1 (Nested Sequence Containment). [thm:nested-sequence-containment] *Let $(A_n)_{n\in\mathbb{N}}$ be a nested sequence of sets. Then*

$$\forall p, q \in \mathbb{N}, \qquad p \leq q \implies A_q \subseteq A_p.$$

Proof. Fix p as any arbitrary natural number. Given that $p \leq q$, it means that $\exists r \in \mathbb{N}$ such that $q = p + r$.
If $r = 0$, then $q = p$ and $A_q = A_p$, which directly implies that $A_q \subseteq A_p$.
If for $r - 1$, if $A_p \subseteq A_{p+r-1}$. We can conclude since $A_{p+r} \subseteq A_{p+r-1}$, we can conclude that $A_{p+r} \subseteq A_{p+r-1} \subseteq A_p$, ie $A_{p+r} \subseteq A_p$.
By Natural Number Induction, we can conclude that $\forall r \in \mathbb{N}, A_{p+r} \subseteq A_p$. Since $q = p + r$, we can conclude that $\forall p, q \in \mathbb{N}, p \leq q \implies A_q \subseteq A_p$. □

Theorem 4.2 (Finite Nested Intersection Property). [thm:finite-nested-intersection] *Let $(A_n)_{n\in\mathbb{N}_1}$ be a sequence of finite, nonempty sets satisfying*

$$\forall n \in \mathbb{N}_1, \quad A_{n+1} \subseteq A_n.$$

Then

$$\emptyset \neq \bigcap_{n\in\mathbb{N}_1} A_n \subseteq A_1.$$

Proof. According to the definition of definition indexed intersection, we want to show that

$$\exists x, \forall n \in \mathbb{N}_1, x \in A_n$$

Given that $\forall n \in \mathbb{N}_1, |(A_n)| \in \mathbb{N}_1$ and $|(A_1)| \geq |(A_2)| \geq \cdots \geq |(A_n)| \geq \cdots$
We define a sequence of natural numbers $(b_n)_{n\in\mathbb{N}_1}$ such that $b_n = |(A_n)| - |(A_{n+1})|$.
For all $m, n \in \mathbb{N}_1$ with $m \leq n$,

$$\begin{aligned}\sum_{k=m}^{n} b_k &= (|A_m| - |A_{m+1}|) + (|A_{m+1}| - |A_{m+2}|) + \cdots + (|A_n| - |A_{n+1}|) \\ &= |A_m| - |A_{n+1}|\end{aligned}$$

Since $A_{n+1} \neq \emptyset$, we have $|A_{n+1}| \geq 1$. Taking $m = 1$,

$$\forall n \in \mathbb{N}_1, \quad \sum_{k=1}^{n} b_k = |A_1| - |A_{n+1}| \leq |A_1| - 1.$$

Since every b_n is a nonnegative integer, only finitely many b_n can be positive. Hence,

$$\exists N_0 \in \mathbb{N}_1, \forall n \geq N_0, b_n = 0.$$

For every $n \geq N_0$,

$$b_n = 0 \implies |A_n| = |A_{n+1}|.$$

Since $A_{n+1} \subseteq A_n$ and A_n is finite, Lemma 1.1 implies that $A_{n+1} = A_n$. Hence

$$\forall n \geq N_0, \quad A_n = A_{N_0}.$$

If $n < N_0$, then $A_{N_0} \subseteq A_n$ by Theorem 4.1. Therefore $A_{N_0} \subseteq A_n$ for every $n \in \mathbb{N}_1$, and thus

$$A_{N_0} \subseteq \bigcap_{n \in \mathbb{N}_1} A_n.$$

The reverse inclusion holds because A_{N_0} is one of the sets in the intersection. Consequently,

$$\bigcap_{n \in \mathbb{N}_1} A_n = A_{N_0},$$

which is finite and nonempty. □

Remark 4.1 (Finiteness is Essential). [rem:finite-nested-intersection-finiteness] The conclusion can fail when the sets are infinite. For example, let

$$A_n = \{m \in \mathbb{N}_1 : m \geq n\}, \qquad n \in \mathbb{N}_1.$$

Every A_n is infinite and nonempty, and

$$A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots, \qquad \bigcap_{n \in \mathbb{N}_1} A_n = \emptyset.$$

Proof. For each $n \in \mathbb{N}_1$, choose $x = n$. Then

$$\forall n \in \mathbb{N}_1, \quad \exists x \in \mathbb{N}_1 : \ x \in A_n.$$

For each $x \in \mathbb{N}_1$, choose $n = x + 1$. Then $x \notin A_n$, so

$$\nexists x \in \mathbb{N}_1, \quad \forall n \in \mathbb{N}_1 : \ x \in A_n.$$

□

5 Intervals

Definition 5.1 (Interval). [def:interval] Let X be a totally ordered set. A set $I \subseteq X$ is an *interval* if, whenever $x, y \in I$ with $x \leq y$, every $z \in X$ satisfying $x \leq z \leq y$ also belongs to I.

Remark 5.1 (Intervals as Sets). [rem:intervals-as-sets] Every interval is a set contained in its ambient totally ordered set X; that is, every interval I satisfies $I \subseteq X$.

Definition 5.2 (Open Interval). [def:open-interval] Let X be a totally ordered set. For $a, b \in X$ with $a < b$, the *open interval* from a to b is

$$(a, b) = \{x \in X : a < x < b\}.$$

Definition 5.3 (Closed Interval). [def:closed-interval] Let X be a totally ordered set. For $a, b \in X$ with $a \leq b$, the *closed interval* from a to b is

$$[a, b] = \{x \in X : a \leq x \leq b\}.$$

Lemma 5.1 (Endpoints of Contained Closed Intervals). [lem:contained-interval-endpoints] *Let $I_1 = [a_1, b_1]$ and $I_2 = [a_2, b_2]$ be closed intervals in the same totally ordered set. If $I_1 \subseteq I_2$, then*

$$a_2 \leq a_1 \leq b_1 \leq b_2.$$

Proof. Since $a_1, b_1 \in I_1 \subseteq I_2$, definition 5.3 gives $a_2 \leq a_1$ and $b_1 \leq b_2$.
Since $I_1 = [a_1, b_1]$ is a closed interval, definition 5.3 requires $a_1 \leq b_1$.
Combining these inequalities gives the result. □

Lemma 5.2 (Endpoint Order of Nested Closed Intervals). [lem:nested-interval-endpoint-order] *Let $I_k = [a_k, b_k]$ be a nested sequence of closed intervals in the same totally ordered set. Then*

$$n \leq m \implies a_n \leq a_m \leq b_m \leq b_n.$$

Proof. **TODO** □

Theorem 5.1 (Nested Intervals Property (NIP)). [thm:nip] *Let F be a complete ordered field, and let $I_n = [a_n, b_n] \subseteq F$ be a sequence of closed intervals.*

$$I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots \implies \bigcap_{n=0}^{\infty} I_n \neq \emptyset.$$

Proof. Let $A = \{a_n : n \in \mathbb{N}\}$. The nesting hypothesis gives

$$\forall n \in \mathbb{N}, \quad I_{n+1} \subseteq I_n.$$

By Theorem 4.1,

$$\forall p, q \in \mathbb{N}, \quad p \leq q \implies I_q \subseteq I_p.$$

By comparability in a total order, $\forall n, m \in \mathbb{N},\ (n \leq m) \vee (m \leq n)$. Hence lemma 5.1 gives

$$\begin{aligned} n \leq m &\implies I_m \subseteq I_n \implies a_n \leq a_m \leq b_m, \\ m \leq n &\implies I_n \subseteq I_m \implies a_n \leq b_n \leq b_m, \end{aligned}$$

so

$$\forall n, m \in \mathbb{N}, \quad a_n \leq b_m.$$

Consequently,

$$a_0 \in A, \forall n \in \mathbb{N},\ a_n \leq b_0 \implies A \neq \emptyset \ \wedge A \text{ is bounded above.}$$

By the completeness of F, choose $\alpha = \sup A$. The definition of supremum gives

$$\forall n, m \in \mathbb{N}, \quad a_n \leq \alpha \leq b_m.$$

Hence, by the definitions of a closed interval and indexed intersection,

$$(\forall n \in \mathbb{N}, \quad a_n \leq \alpha \leq b_n) \implies (\forall n \in \mathbb{N}, \quad \alpha \in I_n) \iff \alpha \in \bigcap_{n=0}^{\infty} I_n.$$

Therefore $\bigcap_{n=0}^{\infty} I_n \neq \emptyset$. □

Corollary 5.1 (Nested Intervals Property in $\mathbb{R}$). [cor:nip-real-numbers] *Let $I_n = [a_n, b_n] \subseteq \mathbb{R}$ be a sequence of closed intervals. Then*

$$I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots \implies \bigcap_{n=0}^{\infty} I_n \neq \emptyset.$$

Proof. By the Axiom of Completeness and the definition of a complete ordered field, $\mathbb{R}$ is complete. The conclusion follows from the Nested Intervals Property. □

Remark 5.2 (Intervals can be Infinitely Small). [rem:inf-small-intervals] If a size of the interval is infinitely small, ie $I = [a, b]$, $\forall \varepsilon > 0$, $b - a < \varepsilon$, this means that the interval is equivalent to a single point, ie $a = b$.

Definition 5.4 (Neighbourhood). [def:neighbourhood] Let F be an ordered field. For $x \in F$ and $\varepsilon \in F$ with $\varepsilon > 0$, the *neighbourhood* of x with radius ε is the open interval

$$V_\varepsilon(x) := (x - \varepsilon,\ x + \varepsilon).$$

6 Decimal Expansion

Theorem 6.1 (Decimal Expansion)**.** [thm:decimal-representation] *Let $D = \{0, 1, \dots, 9\}$. Every digit sequence $a : \mathbb{N}_1 \to D$ determines a unique $x \in [0, 1]$:*

$$\forall a : \mathbb{N}_1 \to D, \quad \exists! x \in [0, 1], \qquad x = \sum_{k=1}^{\infty} a_k 10^{-k}.$$

Proof. We call the summation of the first n decimals s_n, ie

$$s_n = \sum_{k=1}^{n} a_k \, 10^{-k}$$

Since $\forall k \in \mathbb{N}_1, 0 \le a_k \le 9$, for all $n, m \in \mathbb{N}_1$ with $n \le m$,

$$\begin{aligned} s_m &= s_n + \sum_{k=n+1}^{m} a_k 10^{-k} \\ &\le s_n + 9 \sum_{k=n+1}^{m} 10^{-k} \\ &= s_n + 10^{-n} - 10^{-m} < s_n + 10^{-n}. \end{aligned}$$

In other words,

$$\forall n, m \in \mathbb{N}_1, \quad n \le m \implies s_m \in [s_n, s_n + 10^{-n}]$$

By the same argument, we can also show that

$$\forall n, m \in \mathbb{N}_1, \quad n \le m \implies s_m + 10^{-m} \in [s_n, s_n + 10^{-n}]$$

Where we can construct the following nested intervals

$$I_n = [s_n, s_n + 10^{-n}], \qquad I_1 \supseteq I_2 \supseteq \cdots, \qquad x = \sum_{n=\mathbb{N}_1} a_n 10^{-k} = \bigcap_{n \in \mathbb{N}_1} I_n$$

According to the Nested Interval Theorem, after reindexing,

$$\bigcap_{n \in \mathbb{N}_1} I_n \neq \emptyset.$$

Also, notice that $I_n = [s_n, s_n + 10^{-n}]$, $s_n + 10^{-n} - s_n = 10^{-n}$. According to the Remark, we can conclude:

$$\forall \varepsilon > 0, \ \exists n \in \mathbb{N}_1 : \ I_n = [s_n, s_n + 10^{-n}] \text{ and } (s_n + 10^{-n}) - s_n = 10^{-n} < \varepsilon \implies \exists! x \in \bigcap_{n \in \mathbb{N}_1} I_n$$

Thus x is the unique number represented by the digit sequence a. □

Remark 6.1 (Decimal Expansion is not Unique). [rem:decimal-representation-not-unique] The decimal expansion of a real number is not unique. For example, $0.8999\ldots = 0.9000\ldots$

Theorem 6.2 (Uniqueness of Canonical Decimal Expansions)**.** [thm:canonical-decimal-uniqueness] *Let $D = \{0, 1, \dots, 9\}$. For every $x \in [0, 1)$, there exists a unique digit sequence $a : \mathbb{N}_1 \to D$ which is not eventually equal to 9 and satisfies*

$$x = \sum_{n=1}^{\infty} a(n) 10^{-n}.$$

Here, "not eventually equal to 9" means

$$\neg \exists N \in \mathbb{N}_1 \, \forall n \ge N, \quad a(n) = 9.$$

Proof. **TODO** □

7 Archimedean Property

Theorem 7.1 (Archimedean Property). [thm:archimedean-property] *$\mathbb{N}$ is not bounded above in $\mathbb{R}$, in other words, $\forall x \in \mathbb{R}, \exists n \in \mathbb{N}, n > x$*

Proof. Assume for the sake of contradiction that $\mathbb{N}$ is bounded above in $\mathbb{R}$.
Since $0 \in \mathbb{N}$, we have $\mathbb{N} \neq \emptyset$. By the Axiom of Completeness, $\mathbb{N}$ has a supremum, denoted as $\alpha = \sup \mathbb{N}$.
By the definition of epsilon criteria, we have:

$$\forall x \in \mathbb{N}, x \leq \alpha, \quad \forall \varepsilon > 0, \exists x \in \mathbb{N}, \alpha - \varepsilon < x$$

Taking $\varepsilon = 1$, we obtain

$$\exists n \in \mathbb{N} : \quad \alpha - 1 < n \implies \alpha < n + 1.$$

Since $n + 1 = S(n) \in \mathbb{N}$, this contradicts that α is an upper bound of $\mathbb{N}$, namely

$$\forall m \in \mathbb{N}, \quad m \leq \alpha.$$

Therefore, $\mathbb{N}$ is not bounded above in $\mathbb{R}$. □

Theorem 7.2 (Archimedean Property (Reciprocal Form)). [thm:archimedean-reciprocal-form]

$$\forall y > 0, \exists n \in \mathbb{N}_1, 1/n < y$$

Proof. Take $x = 1/y$. Since $y > 0$, we have $x > 0$. By the Archimedean Property, there exists $n \in \mathbb{N}$ with $n > x$. Since $x > 0$, we have $n > 0$, hence $n \in \mathbb{N}_1$. Then $1/n < 1/x = y$. □

Definition 7.1 (Dense Subset). [def:dense-subset] Let $D \subseteq \mathbb{R}$. We say that D is *dense in* $\mathbb{R}$ if

$$\forall a, b \in \mathbb{R}, \quad a < b \implies \exists d \in D : \quad a < d < b.$$

Theorem 7.3 (Density of the Rational Numbers). [thm:rationals-dense-in-reals] *$\mathbb{Q}$ is dense in $\mathbb{R}$.*

Proof. Let $a, b \in \mathbb{R}$ with $a < b$. Then $b - a > 0$, so by the Archimedean property (reciprocal form) there exists $n \in \mathbb{N}_1$ with

$$\frac{1}{n} < b - a \iff 1 < n(b - a) \iff 1 < nb - na.$$

Let $m = \lfloor na \rfloor$, so that $m \leq na < m + 1$.
Set $r := (m + 1)/n \in \mathbb{Q}$.
Notice that

$$na < m + 1 \implies \frac{na}{n} < \frac{m+1}{n} \implies a < r$$

$$1 < nb - na \implies 1 + na < nb \implies 1 + m \leq 1 + na < nb \implies \frac{m+1}{n} < b \implies r < b$$

Therefore $r \in \mathbb{Q} \cap (a, b)$.
So, $\forall a, b \in \mathbb{R}, a < b, \exists r \in \mathbb{Q} : a < r < b$ □

Theorem 7.4 (Density of the Irrational Numbers). [thm:irrationals-dense-in-reals] *$\mathbb{R} \setminus \mathbb{Q}$ (ie, $\mathbb{I}$) is dense in $\mathbb{R}$.*

Proof. **TODO** □

Remark 7.1. $\mathbb{Q}$ and $\mathbb{I}$ are both dense in $\mathbb{R}$. It does not mean that $\mathbb{Q}$ and $\mathbb{I}$ are having the same size. The size of infinite sets are defined in

8 Countability

Definition 8.1 (Countable Infinite Set)**.** [def:countable-infinite-set] A set A is called *countably infinite* if $N \sim A$.

Definition 8.2 (Countable Set)**.** [def:countable-set] A set A is called *countable* if it is either finite or countably infinite.

Theorem 8.1 (Countability of the Rationals)**.** [thm:rationals-natural-equinum] $\mathbb{Q} \sim \mathbb{N}$. *Arrange the positive rationals in the Calkin–Wilf tree, where each a/b has children $a/(a+b)$ and $(a+b)/b$, and read it level by level:*

$$\frac{1}{1}; \qquad \frac{1}{2}, \frac{2}{1}; \qquad \frac{1}{3}, \frac{3}{2}, \frac{2}{3}, \frac{3}{1}; \qquad \dots$$

Thus the levels contain $1, 2, 4, \dots$ elements. Let $r_1, r_2, \dots$ be this enumeration. Since every positive rational occurs exactly once, a witness $f : \mathbb{N} \to \mathbb{Q}$ is

$$f(n) = \begin{cases} 0, & n = 0, \\ r_{(n+1)/2}, & n \text{ is odd}, \\ -r_{n/2}, & n \text{ is even and } n > 0. \end{cases}$$

Theorem 8.2 (Equinumerosity of Intervals)**.** [thm:interval-equinum] *Every interval which is not a single point is equinumerous to another interval which is not a single point.*

Proof. Witness:

$$f : (a, b) \to (c, d), f(x) = \frac{d-c}{b-a}(x-a) + c$$

□

Theorem 8.3 (Intervals and the Real Line)**.** [thm:interval-real-line-equinum] *Every interval which is not a single point is equinumerous to $\mathbb{R}$.*

Proof. For $(-\pi/2, \pi/2)$, the function $f(x) = \tan(x)$ is a bijection from $(-\pi/2, \pi/2)$ to $\mathbb{R}$.
Notice that for every interval (a, b) which is not a single point, we can find a bijection from (a, b) to $(-\pi/2, \pi/2)$ by Theorem 8.2. Since bijections are transitive, we can conclude that every interval (a, b) which is not a single point is equinumerous to $\mathbb{R}$. □

Theorem 8.4 ($\mathbb{Z}, \mathbb{Q}$ are Countable)**.** [thm:integers-rationals-countable] $\mathbb{Z}$ *and* $\mathbb{Q}$ *are countable since they are equinumerous to* $\mathbb{N}$ *according to Theorem 1.3 and Theorem 8.1.*

Theorem 8.5 (Cantor's Theorem)**.** [thm:cantor-theorem] *The interval $(0, 1)$ is uncountable.*

Proof. Since $\mathbb{N} \sim \mathbb{N}_1$, suppose there exists a bijection $f : \mathbb{N}_1 \to (0, 1)$. By Theorem 6.2, choose the unique decimal expansion of each $f(n)$ which is not eventually equal to 9. This defines a function $g : \mathbb{N}_1 \to \mathbb{N}_1 \to \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.
Then, we can define another function of decimal expansion

$$h(n) = \Upsilon(g(n)(n)), \qquad n \in \mathbb{N}_1 \tag{1}$$

$$\Upsilon = \{(0,1), (1,2), (2,3), (3,4), (4,5), (5,6), (6,7), (7,8), (8,1), (9,1)\} \tag{2}$$

Then, we can construct

$$x = \sum_{n=1}^{\infty} h(n) 10^{-n}.$$

Since $h(n) \in \{1, 2, \dots, 8\}$, we have $x \in (0, 1)$, and this is its unique canonical decimal expansion by Theorem 6.2. For every $n \in \mathbb{N}_1$, $h(n) \neq g(n)(n)$, so $x \neq f(n)$. Hence x is not in the image of f, contradicting the assumption that f is a bijection. Therefore, $(0, 1)$ is uncountable. □

Theorem 8.6 ($\mathbb{R}$ is Uncountable)**.** [thm:reals-uncountable] *$\mathbb{R}$ is uncountable.*

Proof. Assume that $\mathbb{R}$ is countable. According to Theorem 8.3, we can conclude that $(0,1) \sim \mathbb{R} \sim \mathbb{N}$. According to Cantor's Theorem, $(0,1)$ is uncountable, which contradicts the assumption that $\mathbb{R}$ is countable. Therefore, $\mathbb{R}$ is uncountable. □

Theorem 8.7 (Countable Union of Countable Sets)**.** [thm:countable-union-of-countable-sets] *Given n countable sets $A_1, A_2, \ldots, A_n$, the union $\bigcup_{i=1}^{n} A_i$ is countable.*

Proof. We can prove this theorem by induction.
Let A and B be two countable sets. If A and B are finite, trivially it is countable. □